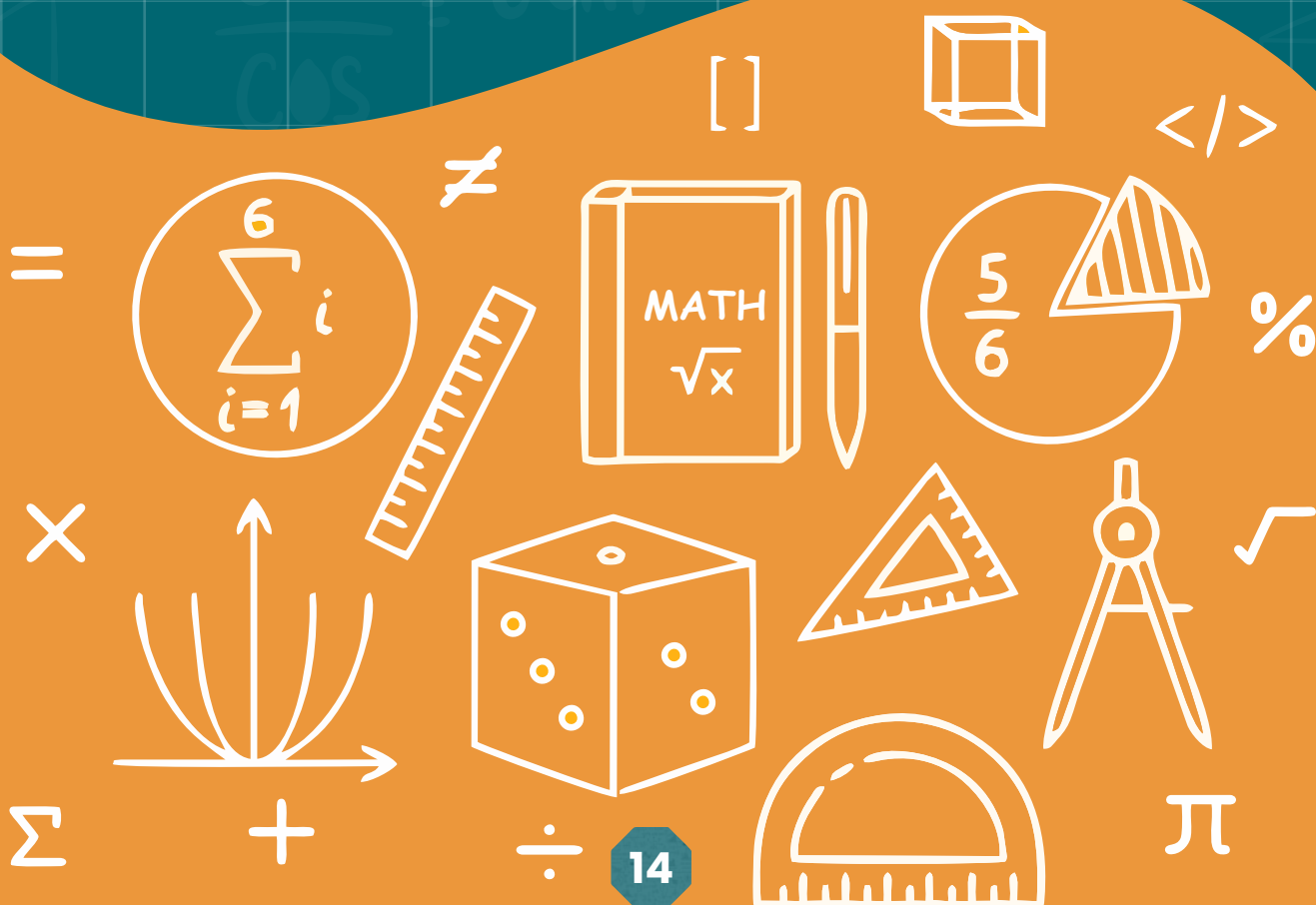


UNIT 2

RATIONAL NUMBERS



RATIONAL NUMBERS

A **rational number** is a number that can be written as a **ratio** (or fraction) of two integers, where the denominator is not zero. The word "rational" comes from the word **ratio**.

Rational numbers can also be represented as decimals.

Examples of Rational Numbers:

1. Fractions: Numbers like $\frac{1}{2}$, $\frac{3}{4}$, and $\frac{6}{5}$ are rational because they are ratios of integers.

2. Integers: Numbers like -2 , 0 , and 3 are also rational because they can be written as fractions $-\frac{2}{1}$ and $\frac{3}{1}$.

3. Terminating Decimals: Decimals that stop, like 0.35 , 0.7116 , and 0.9768 , can be converted into fractions.

4. Non-Terminating Decimals with Repeating Patterns:

Decimals like $0.333\ldots$ (repeating 3) and $0.141414\ldots$ (repeating 14) are rational because they follow a repeating pattern and can be written as fractions.

The table given below shows examples which will help you in identifying terminating decimals better.

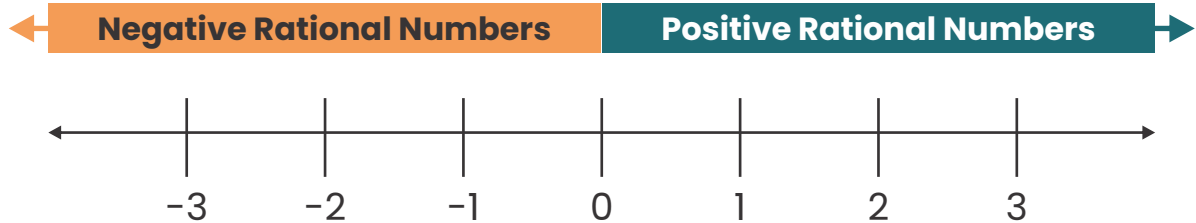
Number	Terminating or Non-terminating Decimal
2.675	Since there are 3 digits after the decimal point, it is a terminating decimal number.
$\frac{3}{8}$	We can write $\frac{3}{8}$ as 0.375 . So, it is a terminating decimal number.
$\frac{1}{6}$	If we convert it into decimal, the answer will be $0.16666\ldots$. Since 6 keeps on repeating, it is a non-terminating decimal with repeating patterns.

NUMBER LINE

The number line is a visual representation of numbers arranged in increasing order from left to right.

Some examples include: $\frac{3}{4}$, 5, -125, and $-\frac{1}{2}$

All of these numbers can be represented on the number line.



All negative rational numbers are to the left of zero on the number line. In comparison, all positive rational numbers are to the right of zero on the number line.

OPERATION ON RATIONAL NUMBERS

Addition of Rational Numbers

Case I. Rational numbers having the same denominator

To add two rational numbers with the same denominators, we add their numerators; the denominator remains the same.

In other words, if $\frac{p}{q}$ and $\frac{r}{q}$ ($q > 0$) are two rational numbers, then

$$\frac{p}{q} + \frac{r}{q} = \frac{p+r}{q}$$

Example: Add $\frac{-7}{11}$ and $\frac{3}{11}$

Solution:

First expressing $\frac{-7}{11}$ as a rational number with a positive denominator, we have

$$\frac{-7}{11} = \frac{7 \times (-1)}{-11 \times (-1)} = \frac{-7}{11}$$

Now, the denominators are the same, we can go ahead and add the numerators

$$\frac{-7}{11} + \frac{3}{11} = \frac{-7}{11} + \frac{3}{11} = \frac{(-7) + 3}{11} = \frac{-4}{11}$$

Case II. Rational numbers having different denominators

To add two rational numbers to be added do not have common denominators, find LCM of their denominators and express each rational number with this LCM as their denominator, then add them as in Case I.

Example: Add $\frac{-3}{5}$ and $\frac{2}{3}$

First, we find the LCM of the denominators 5 and 3.

$$\text{LCM}(5, 3) = 15$$

Expressing each rational number with this LCM i.e., 15 as their denominator, we have

$$\frac{-3}{5} = \frac{(-3) \times 3}{5 \times 3} = \frac{-9}{15} \quad \text{and} \quad \frac{2}{3} = \frac{2 \times 5}{3 \times 5} = \frac{10}{15}$$

$$\begin{aligned} \therefore \frac{-3}{5} + \frac{2}{3} &= \frac{-9}{15} + \frac{10}{15} \\ &= \frac{(-9) + 10}{15} \\ &= \frac{1}{15} \end{aligned}$$

Subtraction:

If $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then subtracting $\frac{c}{d}$ from $\frac{a}{b}$ means adding the additive inverse of $\frac{c}{d}$ to $\frac{a}{b}$ i.e.,

$$\frac{a}{b} - \frac{c}{d} = \frac{a}{b} + \left(-\frac{c}{d}\right)$$

Example:

Subtract $\frac{5}{7}$ from $\frac{-3}{14}$

$$\begin{aligned} \frac{-3}{14} - \frac{5}{7} &= \frac{-3}{14} + \frac{-5}{7} &= \frac{-3 + (-10)}{14} \\ &= \frac{-3}{14} + \frac{-10}{14} &= \frac{-13}{14} \end{aligned}$$

Example:

Evaluate $\frac{3}{5} + \frac{10}{3} - \frac{11}{15}$

First, we find the LCM of the denominators 5, 3, 15 .

$$\text{LCM}(5, 3, 15) = 15$$

Expressing each rational number with this LCM i.e., 15 as their denominator, we have

$$\begin{aligned} \frac{3}{5} + \frac{10}{3} - \frac{11}{15} &= \frac{(3 \times 3) + (10 \times 5) - 11}{15} \\ &= \frac{9 + 50 - 11}{15} \\ &= \frac{48}{15} \end{aligned}$$

Multiplication:

Multiplication of rational numbers is done in the same way as multiplication of fractions. Thus,

$$\text{Product of Rational numbers} = \frac{\text{Product of numerators}}{\text{Product of denominators}}$$

In other words, if $\frac{a}{b}$ and $\frac{c}{d}$ are two rational numbers, then

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Example:

Multiply $\frac{6}{11}$ by $\frac{-2}{3}$

$$\begin{aligned} \frac{6}{11} \times \frac{-2}{3} &= \frac{-12}{33} \\ &= \frac{-4}{11} \end{aligned}$$

Division:

The division of fractions is the inverse of multiplication

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \left(\frac{c}{d}\right)^{-1} = \frac{a}{b} \times \frac{d}{c}$$

Example:

Divide $\frac{-6}{11}$ by $\frac{5}{7}$

$$\begin{aligned} \frac{-6}{11} \div \frac{5}{7} &= \frac{-6}{11} \times \frac{7}{5} \quad (\text{Multiply by the reciprocal of } \frac{5}{7}) \\ &= \frac{-6 \times 7}{11 \times 5} = \frac{-42}{55} \end{aligned}$$

Comparing Fractions:

We can compare two positive fractions by cross multiplying their numerators and denominators. That fraction will be greater whose numerator produces the greater product. For example, if we want to find which is greater among $\frac{a}{b}$ and $\frac{c}{d}$, we will calculate $a \times d$ and $b \times c$.

If $a \times d > b \times c$, then $\frac{a}{b} > \frac{c}{d}$

If $a \times d < b \times c$, then $\frac{c}{d} > \frac{a}{b}$

If $a \times d = b \times c$, then $\frac{a}{b} = \frac{c}{d}$

EXERCISE 2.1

Q1. Put the correct sign $>$ or $<$ between the following pair of rational numbers:

i. $\frac{2}{3}, \frac{1}{5}$

ii. $\frac{10}{13}, \frac{11}{14}$

iii. $\frac{1}{4}, \frac{1}{2}$

iv. $\frac{11}{17}, \frac{3}{8}$

v. $\frac{4}{9}, \frac{2}{5}$

vi. $\frac{5}{22}, \frac{11}{25}$

Q2. Find the sum:

i. $\frac{-16}{3} + \frac{7}{3}$

ii. $\frac{-2}{3} + \frac{3}{7}$

iii. $\frac{3}{11} + \frac{7}{-15}$

iv. $\frac{-9}{24} + \frac{-5}{18}$

v. $\frac{29}{-4} + \frac{-15}{8}$

vi. $\frac{-9}{7} + \frac{3}{7} + \frac{-6}{7}$

Q3. Evaluate:

i. $\frac{-3}{11} - \frac{-5}{11}$

ii. $0 - \frac{-9}{12}$

iii. $\frac{-3}{8} - (-5)$

iv. $1 - \frac{-11}{3}$

v. $\frac{-16}{3} + \frac{7}{3}$

vi. $\frac{-2}{3} + \frac{3}{7}$

vii. $\frac{-3}{11} + \frac{7}{-15}$

viii. $\frac{-9}{24} + \frac{-5}{18}$

ix. $\frac{29}{-4} + \frac{-15}{18}$

x. $\frac{-9}{7} + \frac{3}{7} + \frac{-6}{7}$

Q4. Multiply the following:

i. $\frac{-1}{25}$ by $\frac{2}{24}$

ii. -1 by $\frac{14}{56}$

iii. $\frac{-3}{10}$ by $\frac{-4}{21}$

iv. -36 by $\frac{-5}{9}$

v. $\frac{-7}{9}$ by $\frac{-9}{8}$

Q5. Divide the following:

i. $\frac{-7}{13} \div 26$

ii. $\frac{-3}{5} \div (-9)$

iii. $14 \div \frac{-2}{3}$

iv. $\frac{-7}{9} \div \frac{-21}{-26}$

Q6. Solve the following.

i. $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$

ii. $\frac{2}{7} + \frac{1}{2} - \frac{3}{14}$

iii. $\frac{5}{8} - \frac{1}{5} - \frac{3}{4}$

iv. $\left(-\frac{1}{3}\right) + \left(-\frac{1}{4}\right) + \frac{1}{2}$

v. $\left(-\frac{11}{15}\right) + \left(-\frac{3}{5}\right) + \frac{5}{4}$

Q9. Simplify the following.

i. $\left(-\frac{1}{100}\right) \div \left(-\frac{1}{10}\right)$

ii. $\frac{1}{5} \times \frac{2}{3} \times \left(-\frac{30}{44}\right)$

iii. $\frac{2}{3} \div \frac{16}{21} - \frac{27}{49}$

iv. $\frac{8}{21} \div \frac{7}{12}$

v. $\frac{1}{6} + \left(-\frac{2}{3}\right) + \left(-\frac{11}{63}\right)$

vi. $-\frac{2}{7} \div \frac{3}{4} \times \frac{63}{100}$

PROPERTIES

Property	Operations (Here Z means integers e.g. 0, ±1, ±2...)			
	Addition	Subtraction	Multiplication	Division
Closure	$a + b \in Z$	$a - b \in Z$	$a \times b \in Z$	$a \div b \notin Z$
Commutative	$a + b = b + a$	$a - b \neq b - a$	$a \times b = b \times a$	$a \div b \neq b \div a$
Associative	$a + (b + c) = (a + b) + c$	$a - (b - c) \neq (a - b) - c$	$a \times (b \times c) = (a \times b) \times c$	$a \div (b \div c) \neq (a \div b) \div c$
Distributive	$a \times (b + c) = ab + ac$	$a \times (b - c) = ab - ac$	Not Applicable	Not Applicable
Identity	$a + 0 = 0 + a = a$	Not Applicable	$a \times 1 = 1 \times a = a$	Not Applicable

Associative property w.r.t addition:

We know that the sum of any three rational numbers in any order remains same, i.e., for three rational numbers a/b , c/d , and e/f :

$$\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right) = \left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f}$$

This property of rational numbers is called associative property w.r.t addition.

Example:

Verify associative property w.r.t addition for $\frac{1}{3}$, $\frac{2}{5}$, and $\frac{3}{4}$.

Solution: According to associative property w.r.t addition:

$$\frac{1}{3} + \left(\frac{2}{5} + \frac{3}{4} \right) = \left(\frac{1}{3} + \frac{2}{5} \right) + \frac{3}{4}$$

Now,

$$\begin{aligned} \text{LHS} &= \frac{1}{3} + \left(\frac{2}{5} + \frac{3}{4} \right) \\ &= \frac{1}{3} + \left(\frac{8+15}{20} \right) \\ &= \frac{1}{3} + \frac{23}{20} \\ &= \frac{20+69}{60} = \frac{89}{60} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \left(\frac{1}{3} + \frac{2}{5} \right) + \frac{3}{4} \\ &= \left(\frac{5+6}{15} \right) + \frac{3}{4} \\ &= \frac{11}{15} + \frac{3}{4} \\ &= \frac{44+45}{60} = \frac{89}{60} \end{aligned}$$

Since,

LHS = RHS

Therefore, $\frac{1}{3} + \left(\frac{2}{5} + \frac{3}{4} \right) = \left(\frac{1}{3} + \frac{2}{5} \right) + \frac{3}{4}$

Hence verified.

We know that the sum of any three rational numbers in any order remains same, i.e for three rational numbers $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$

$$\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f} \right) = \left(\frac{a}{b} + \frac{c}{d} \right) + \frac{e}{f}$$

This property of rational numbers is called associative property w.r.t addition.

Associative property w.r.t multiplication:

We know that the product of any three rational numbers $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$ in any order remains the same.

i.e.,
$$\frac{a}{b} \times \left(\frac{c}{d} \times \frac{e}{f} \right) = \left(\frac{a}{b} \times \frac{c}{d} \right) \times \frac{e}{f}$$

This property of rational numbers is called associative property w.r.t multiplication.

Example:

Verify associative property w.r.t multiplication for $\frac{1}{3}$, $\frac{2}{5}$, and $\frac{3}{4}$.

Solution: According to associative property w.r.t multiplication:

$$\frac{1}{3} \times \left(\frac{2}{5} \times \frac{3}{4} \right) = \left(\frac{1}{3} \times \frac{2}{5} \right) \times \frac{3}{4}$$

Now,

$$\begin{aligned} \text{LHS} &= \frac{1}{3} \times \left(\frac{2}{5} \times \frac{3}{4} \right) \\ &= \frac{1}{3} \times \left(\frac{2 \times 3}{5 \times 4} \right) \\ &= \frac{1}{3} \times \frac{6}{20} \\ &= \frac{6}{60} = \frac{1}{10} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \left(\frac{1}{3} \times \frac{2}{5} \right) \times \frac{3}{4} \\ &= \left(\frac{1 \times 2}{3 \times 5} \right) \times \frac{3}{4} \\ &= \frac{2}{15} \times \frac{3}{4} \\ &= \frac{6}{60} = \frac{1}{10} \end{aligned}$$

Since,

LHS = RHS

Therefore,
$$\frac{1}{3} \times \left(\frac{2}{5} \times \frac{3}{4} \right) = \left(\frac{1}{3} \times \frac{2}{5} \right) \times \frac{3}{4}$$

Hence verified.

We know that the product of any three rational numbers in any order remains same, i.e for three rational numbers $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$

$$\frac{a}{b} \times \left(\frac{c}{d} \times \frac{e}{f} \right) = \left(\frac{a}{b} \times \frac{c}{d} \right) \times \frac{e}{f}$$

Distributive Property of Multiplication Over Addition

The distributive property of multiplication over addition is applied when we need to multiply a number by the sum of two numbers i.e. For $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$ are any three rational numbers.

$$\frac{a}{b} \times \left(\frac{c}{d} + \frac{e}{f} \right) = \frac{a}{b} \times \frac{c}{d} + \frac{a}{b} \times \frac{e}{f}$$

This property of rational numbers is called associative property w.r.t multiplication.

Example:

Verify distributive property of rational numbers w.r.t. multiplication over addition, for $\frac{1}{7}$, $\frac{2}{7}$ and $\frac{3}{7}$.

Solution: According to associative property w.r.t multiplication over addition: $\frac{1}{7} \times \left(\frac{2}{7} + \frac{3}{7} \right) = \frac{1}{7} \times \frac{2}{7} + \frac{1}{7} \times \frac{3}{7}$

Now,

$$\begin{aligned} \text{LHS} &= \frac{1}{7} \times \left(\frac{2}{7} + \frac{3}{7} \right) \\ &= \frac{1}{7} \times \left(\frac{2+3}{7} \right) \\ &= \frac{1}{7} \times \frac{5}{7} \\ &= \frac{5}{49} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \frac{1}{7} \times \frac{2}{7} + \frac{1}{7} \times \frac{3}{7} \\ &= \frac{1 \times 2}{7 \times 7} + \frac{1 \times 3}{7 \times 7} \\ &= \frac{2}{49} + \frac{3}{49} \\ &= \frac{2+3}{49} = \frac{5}{49} \end{aligned}$$

Since,

LHS = RHS

$$\text{Therefore, } \frac{1}{7} \times \left(\frac{2}{7} + \frac{3}{7} \right) = \frac{1}{7} \times \frac{2}{7} + \frac{1}{7} \times \frac{3}{7}$$

Hence verified.

Distributive Property of Multiplication Over Subtraction

The distributive property of multiplication over subtraction is similar to the distributive property of multiplication over addition except for the operation of addition and subtraction.

i.e. For $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$ are any three rational numbers.

$$\frac{a}{b} \times \left(\frac{c}{d} - \frac{e}{f} \right) = \frac{a}{b} \times \frac{c}{d} - \frac{a}{b} \times \frac{e}{f}$$

This property of rational numbers is called associative property w.r.t multiplication.

Example:

Verify distributive property of rational numbers w.r.t. multiplication over subtraction, for $\frac{4}{5}$, $\frac{3}{5}$, and $\frac{2}{5}$.

Solution: According to associative property w.r.t multiplication over subtraction:

$$\frac{4}{5} \times \left(\frac{3}{5} - \frac{2}{5} \right) = \frac{4}{5} \times \frac{3}{5} - \frac{4}{5} \times \frac{2}{5}$$

Now,

$$\begin{aligned} \text{LHS} &= \frac{4}{5} \times \left(\frac{3}{5} - \frac{2}{5} \right) \\ &= \frac{4}{5} \times \left(\frac{3-2}{5} \right) \\ &= \frac{4}{5} \times \frac{1}{5} \\ &= \frac{4}{25} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \frac{4}{5} \times \frac{3}{5} - \frac{4}{5} \times \frac{2}{5} \\ &= \frac{4 \times 3}{5 \times 5} - \frac{4 \times 2}{5 \times 5} \\ &= \frac{12}{25} - \frac{8}{25} \\ &= \frac{12-8}{25} = \frac{4}{25} \end{aligned}$$

Since,

LHS = RHS

$$\text{Therefore, } \frac{4}{5} \times \left(\frac{3}{5} - \frac{2}{5} \right) = \frac{4}{5} \times \frac{3}{5} - \frac{4}{5} \times \frac{2}{5}$$

Hence verified.

EXERCISE 2.2

Q1. Verify associative property w.r.t addition for $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$.

Q2. Verify associative property w.r.t multiplication for $\frac{2}{7}$, $\frac{4}{5}$ and $\frac{3}{4}$.

Q3. Verify distributive property of multiplication over addition for $\frac{3}{4}$, $\frac{4}{5}$ and $\frac{7}{8}$.

Q4. Verify distributive property of multiplication over subtraction for $\frac{5}{7}$, $\frac{3}{4}$ and $\frac{1}{5}$.

Q5. Name the properties used in the following.

i. $\left(\frac{-1}{2}\right) + \frac{1}{3} = \frac{1}{3} + \left(\frac{-1}{2}\right)$

ii. $\frac{-10}{11} \times \left(\frac{-5}{44}\right) = \left(\frac{-5}{44}\right) \times \left(\frac{-10}{11}\right)$

iii. $\frac{-2}{3} \times \left(\frac{7}{8} \times \frac{9}{14}\right) = \left(\frac{-2}{3} \times \frac{7}{8}\right) \times \frac{9}{14}$ iv. $\frac{-3}{5} + \left(\frac{1}{2} + \frac{7}{10}\right) = \left(\frac{-3}{5} + \frac{1}{2}\right) + \frac{7}{10}$

v. $\frac{-1}{2} + \left(\frac{3}{5} + \frac{1}{4}\right) = \left(\frac{-1}{2} + \frac{3}{5}\right) + \frac{1}{4}$

vi. $\frac{2}{3} \times \left(\frac{1}{2} + \frac{5}{6}\right) = \left(\frac{2}{3} \times \frac{1}{2}\right) + \left(\frac{2}{3} \times \frac{5}{6}\right)$

vii. $\left(\frac{-12}{105}\right) \times \left(\frac{-15}{84}\right) = \left(\frac{-15}{84}\right) \times \left(\frac{-12}{105}\right)$

viii. $\frac{1}{4} \times \left(\frac{8}{9} - \frac{12}{15}\right) = \left(\frac{1}{4} \times \frac{8}{9}\right) - \left(\frac{1}{4} \times \frac{12}{15}\right)$

ix. $\frac{-5}{8} \times \left(\frac{4}{7} - \frac{2}{3}\right) = \left(\frac{-5}{8} \times \frac{4}{7}\right) - \left(\frac{-5}{8} \times \frac{2}{3}\right)$

x. $\frac{24}{49} \times \left(\frac{7}{8} + \frac{14}{6}\right) = \left(\frac{24}{49} \times \frac{7}{8}\right) + \left(\frac{24}{49} \times \frac{14}{6}\right)$

Q6. Prove that.

i. $(-1) + \frac{35}{54} = \frac{35}{54} + (-1)$

ii. $\left(\frac{-121}{169}\right) \times \left(\frac{-13}{11}\right) = \left(\frac{-13}{11}\right) \times \left(\frac{-121}{169}\right)$

iii. $\frac{4}{9} \times \left(\frac{2}{3} \times \frac{5}{7}\right) = \left(\frac{4}{9} \times \frac{2}{3}\right) \times \frac{5}{7}$

iv. $\frac{-4}{5} \times \left(\frac{1}{8} + \frac{11}{12}\right) = \left(\frac{-4}{5} \times \frac{1}{8}\right) + \left(\frac{-4}{5} \times \frac{11}{12}\right)$

v. $\frac{-1}{4} + \left(\frac{1}{6} + \frac{3}{5}\right) = \left(\frac{-1}{4} + \frac{1}{6}\right) + \frac{3}{5}$

iv. $\frac{5}{12} \times \left(\frac{-2}{7} - 2\right) = \left(\frac{5}{12} \times \frac{-2}{7}\right) - \left(\frac{5}{12} \times 2\right)$